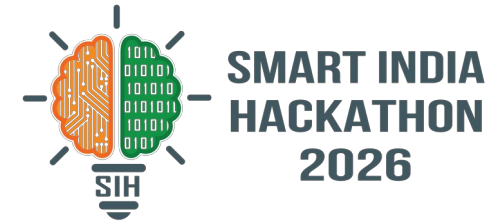


# SMART INDIA HACKATHON 2026



## TITLE PAGE

**Problem Statement ID: SIH26183**

### **Problem Statement Title**

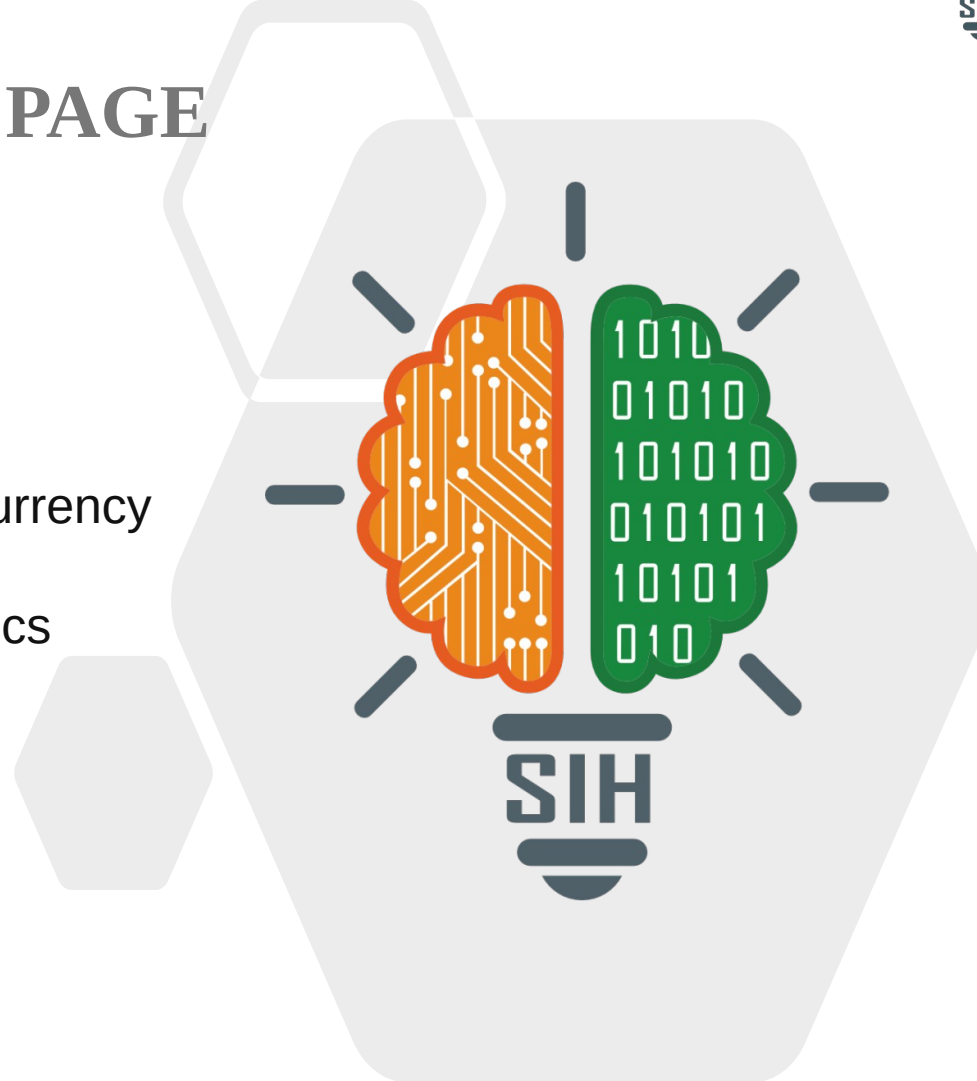
Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

**Theme: Blockchain & Cybersecurity**

**PS Category: Software**

**Team ID: [ENTER REGISTERED TEAM ID]**

**Team Name (Registered on portal):  
COLD COFFEE**



## Proposed Solution (Describe your Idea/Solution/Prototype)

### Detailed explanation of the proposed solution

- Ingest victim-reported addresses and trace outgoing funds through intermediary wallets.
- Identify the nearest supported receiving exchange/VASP with its deposit transaction and path.
- Monitor subsequent movement and generate alerts with standardized investigation reports.

### How it addresses the problem

- Give investigators transaction hashes, amounts, timestamps and sourced attribution evidence.
- Recommend follow-up actions; plan NCRP/SAHYOG integration through approved interfaces.

### Innovation and uniqueness of the solution

- Combine receiving-VASP discovery with evidence provenance and movement monitoring.
- Keep observed facts, inferred links and unresolved paths distinct for investigator review.

## Technologies to be used

- React dashboard; Python/FastAPI services; queued workers and persistent case storage.
- Bitcoin, EVM and TRON adapters; incremental indexing and versioned VASP address labels.
- Bounded graph search, cautious clustering, explainable rules and evaluated ML risk detection.

## Methodology and process for implementation



Cross-chain correlation where supported; uncertainty remains explicit

Planned integration: complaint APIs and NCRP/SAHYOG, subject to approved access.

## Analysis of the feasibility of the idea

- Prototype foundation: bounded Bitcoin tracing, incoming/outgoing graph views and PDF/CSV export.
- Planned endpoint: multi-hop VASP discovery, monitoring, additional chains and LEA integration.

## Potential challenges and risks

- Rate limits and large histories; incomplete or outdated exchange-address labels.
- Mixing activity and ambiguous ownership; unsupported bridges and restricted integration access.

## Strategies for overcoming these challenges

- Use indexing, caching and bounded traversal; retain label sources and mark tracing gaps.
- Validate precision, coverage, false positives and latency before expanding the deployment.

## Potential impact on the target audience

- Cybercrime investigators: prioritize cases with identifiable receiving VASPs and track new movement.
- Investigators and prosecutors: review transaction paths, attribution sources and analysis records.
- Financial intelligence units: triage activity using explainable patterns and evidence-linked indicators.

## Benefits of the solution (social, economic, environmental, etc.)

- Social: support faster preservation and freezing requests through investigator-reviewed findings.
- Economic: reduce repetitive tracing effort with open-source components and shared infrastructure.
- Operational: standardize evidence reports and improve coordination with identified VASPs.

Expected benefits to be measured in pilots; fund recovery and attribution are not guaranteed.

## Details / Links of the reference and research work

|                             |                                                                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Address clustering</b>   | Meiklejohn et al. (2013), A Fistful of Bitcoins: Characterizing Payments Among Men with No Names.<br><a href="#">ACM IMC paper</a>                                   |
| <b>ML-assisted analysis</b> | Weber et al. (2019), Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics.<br><a href="#">arXiv:1908.02591</a>  |
| <b>Blockchain data</b>      | Blockstream Esplora: address and transaction API documentation.<br><br><a href="#">Official API documentation</a>                                                    |
| <b>Attribution sources</b>  | Binance, Our Commitment to Transparency (2022): published wallet disclosures.<br><br><a href="#">Historical disclosure; not a deposit-account registry</a>           |
| <b>AML context</b>          | FATF (2021), Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs.<br><br><a href="#">FATF guidance</a>                                            |
| <b>Evidence handling</b>    | NIST SP 800-86 (2006), Guide to Integrating Forensic Techniques into Incident Response.<br><br><a href="#">NIST guidance; not a guarantee of legal admissibility</a> |